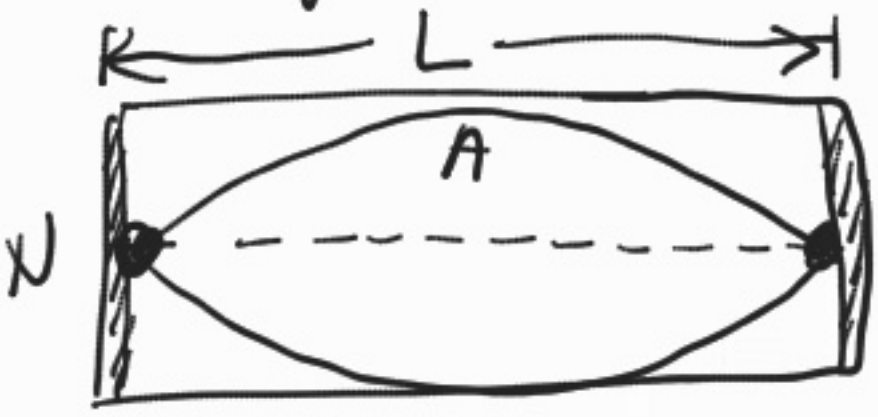
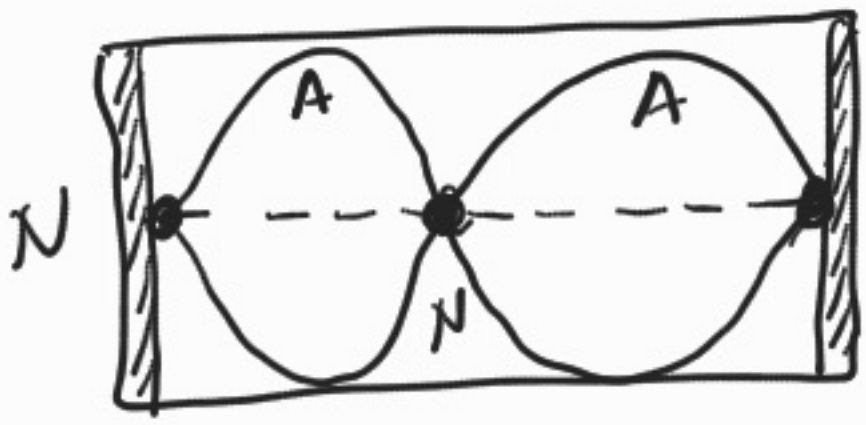
Standing Sound Waves

- Pipe closed on both ends.
- Displacement Profile.



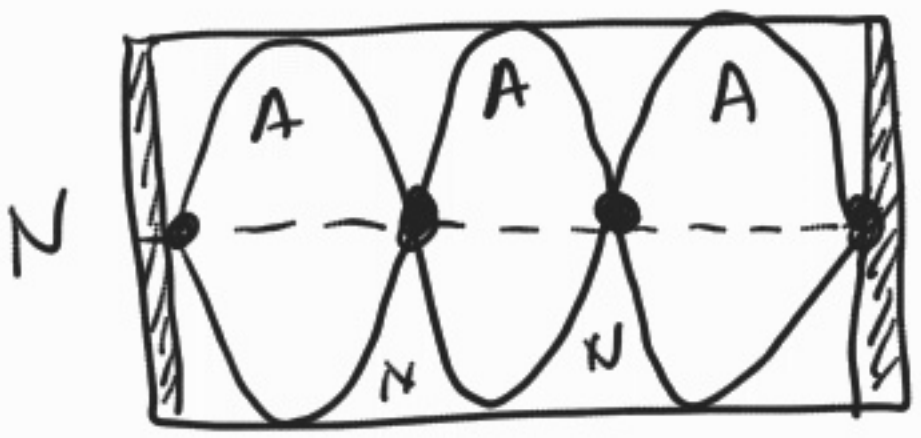
1st Harmonic

$\lambda_1 = 2L$



2nd Harmonic

$\lambda_2 = L$



3rd Harmonic

$\lambda_3 = \frac{2L}{3}$

- frequency of the nth harmonic (normal mode frequencies):

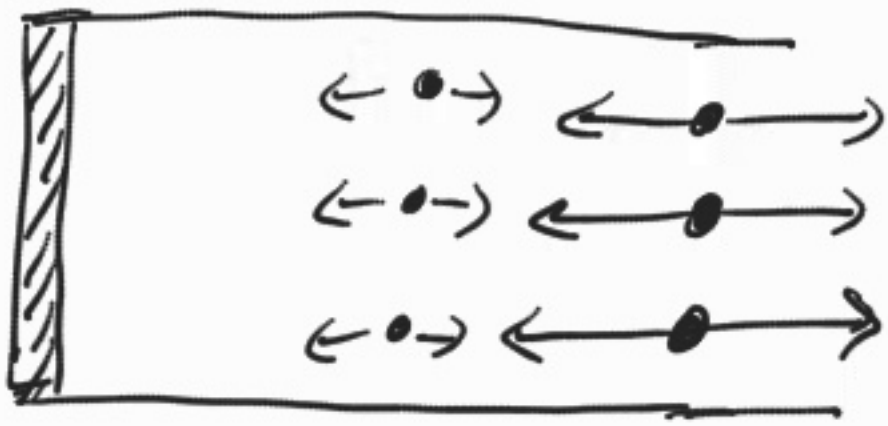
$f_n = nv/2L$ $\quad \lambda_n = \frac{2L}{n}$; $n = 1, 2, 3, \ldots$

$f_1 = v/2L$

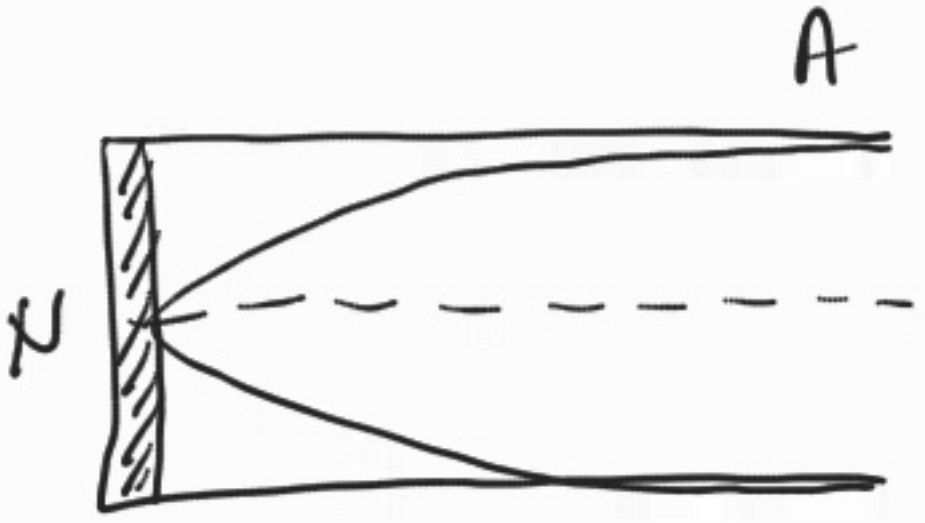
(fundamental f)

Standing Sound Waves

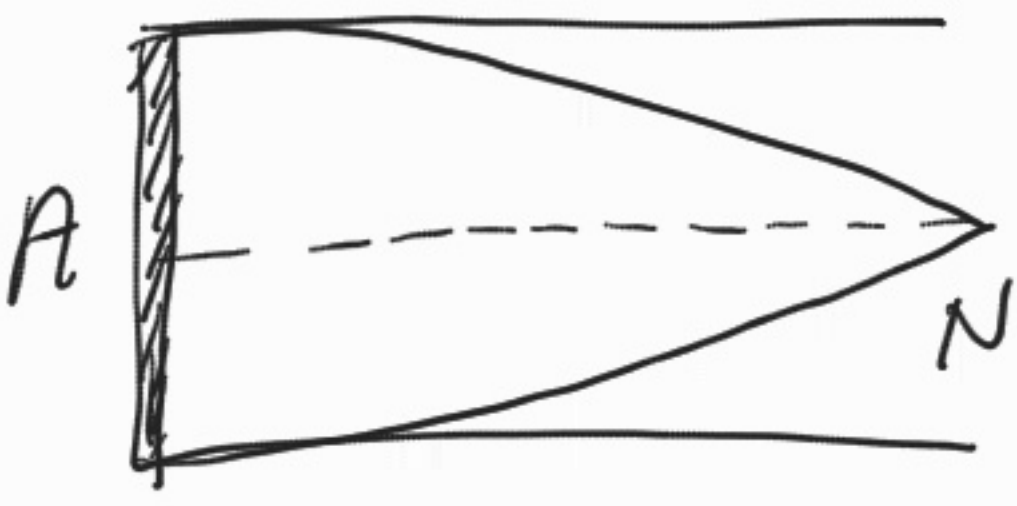
- A pipe with one end closed
- 1st Harmonic



} molecules of air longitudinally oscillate.



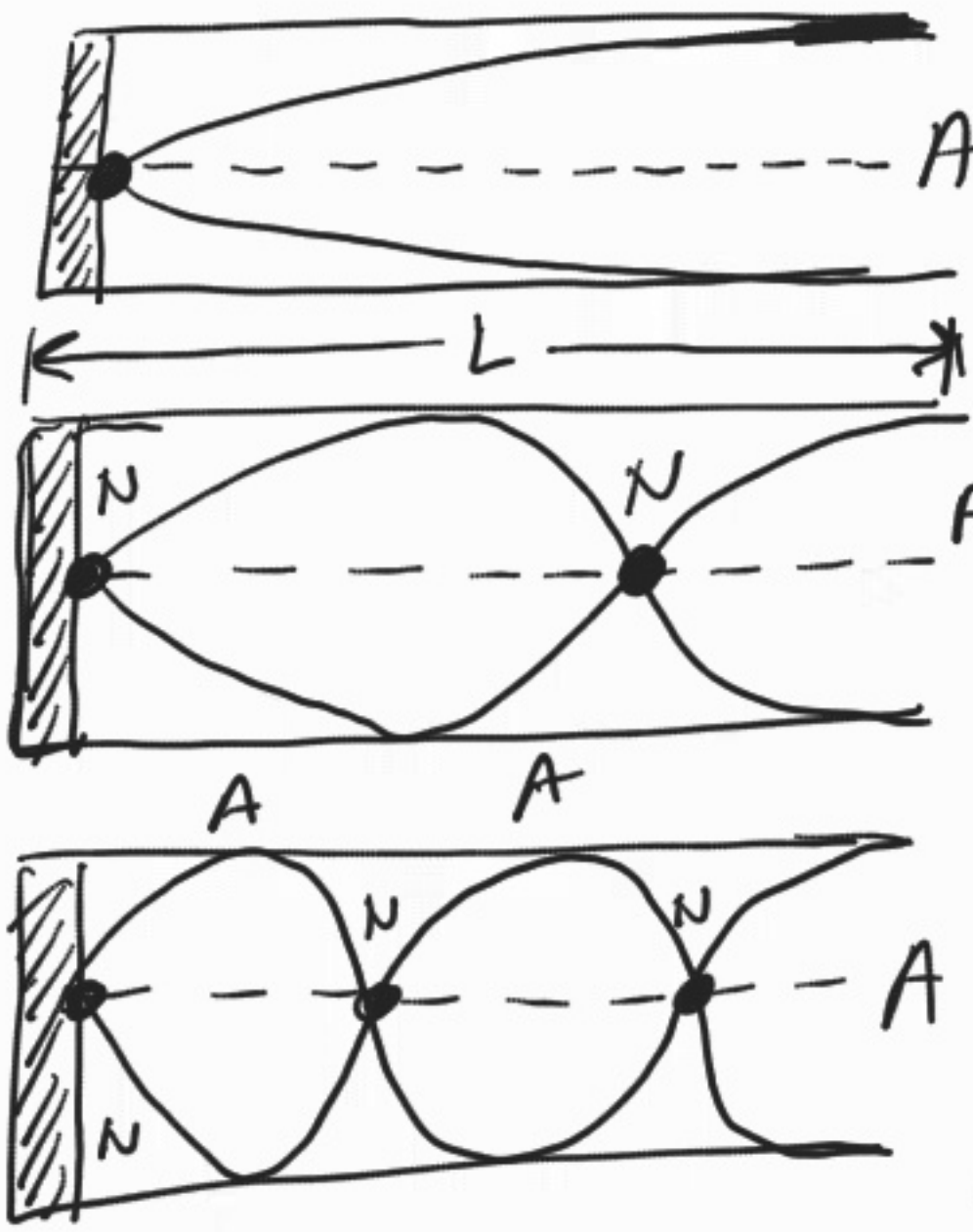
} displacement at closed end is zero & max at open end.



} pressure at the closed end is at max & zero at open end.

Standing Sound Waves

- pipe with one end closed.
- displacement profile.



$\lambda_1 = 4L$
[1st Harmonic]

$\lambda_3 = 4L/3$
2nd Harmonic]

$\lambda_5 = 4L/5$
[3rd Harmonic]

- frequency of the nth harmonic (normal mode f)

$f_n = nv/4L$

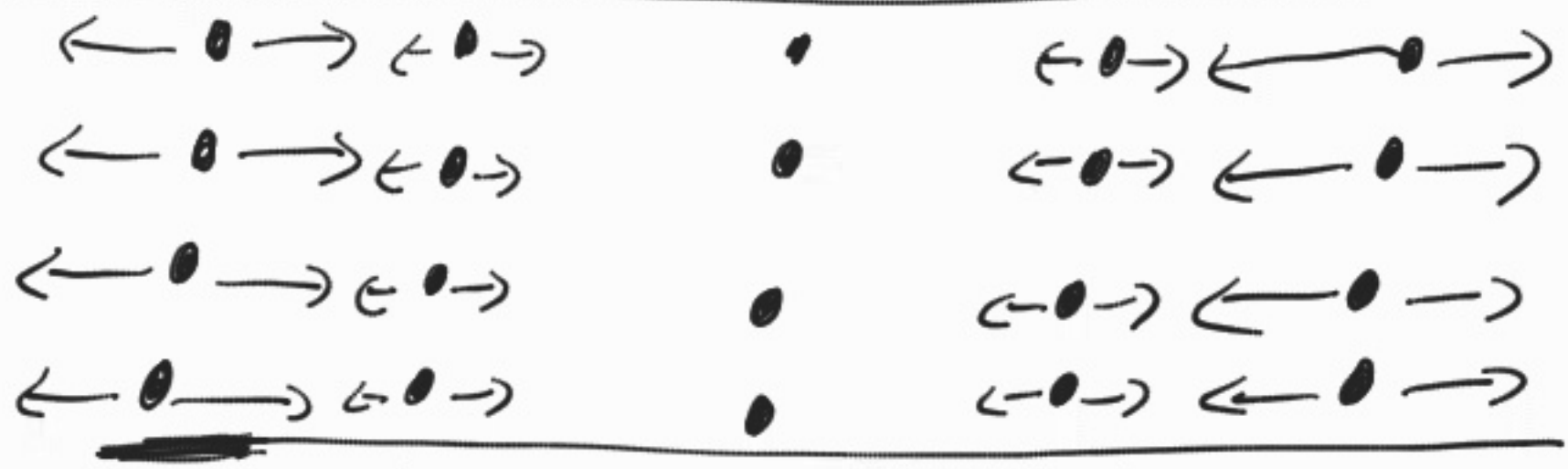
$\lambda_n = 4L/n$

where
$n = 1, 2, 3, \cdots$

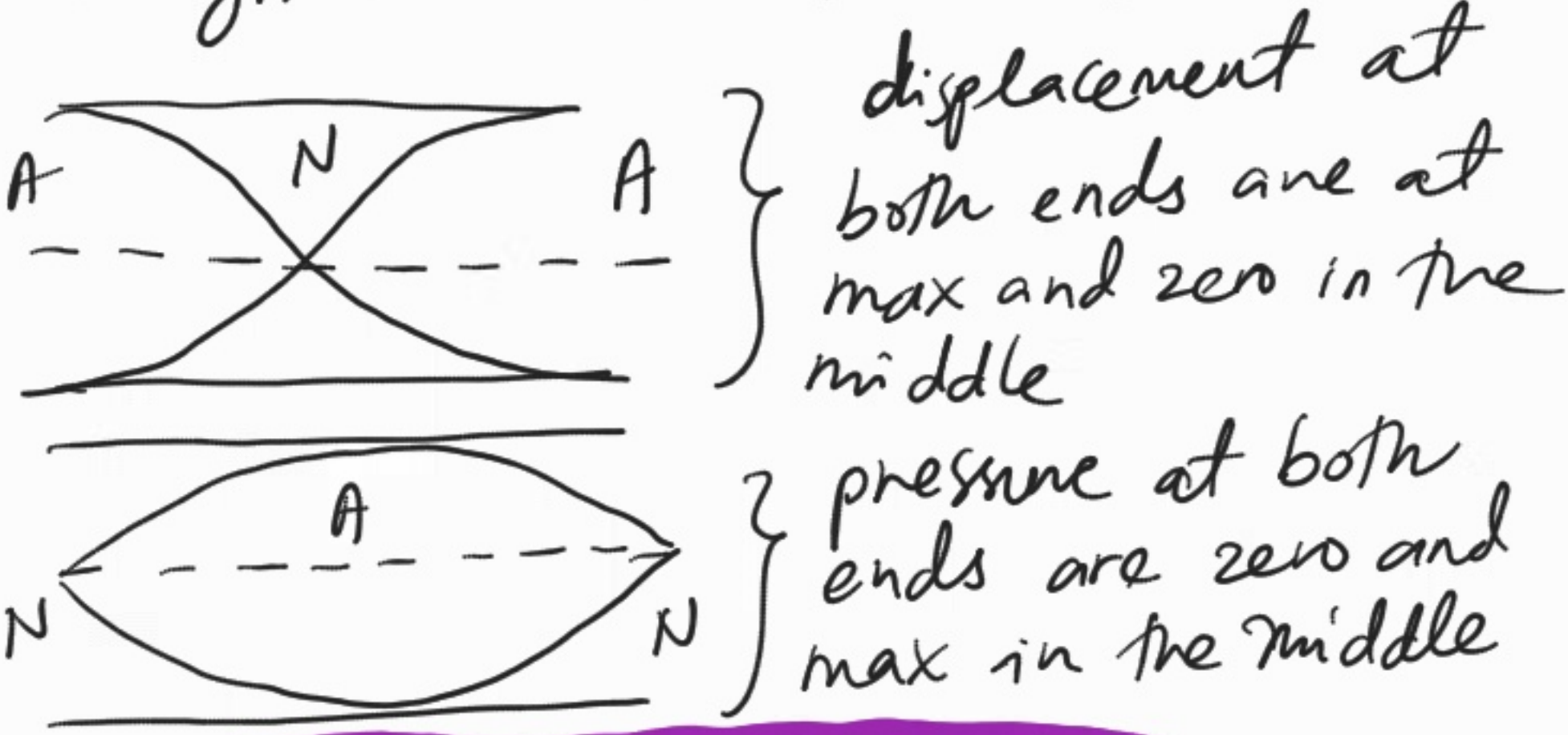
[$f_1 = v/4L$
(fundamental f)

Standing Sound Waves

- A pipe with both ends open
- 1st Harmonic



#molecules of air undergo longitudinal oscillations.

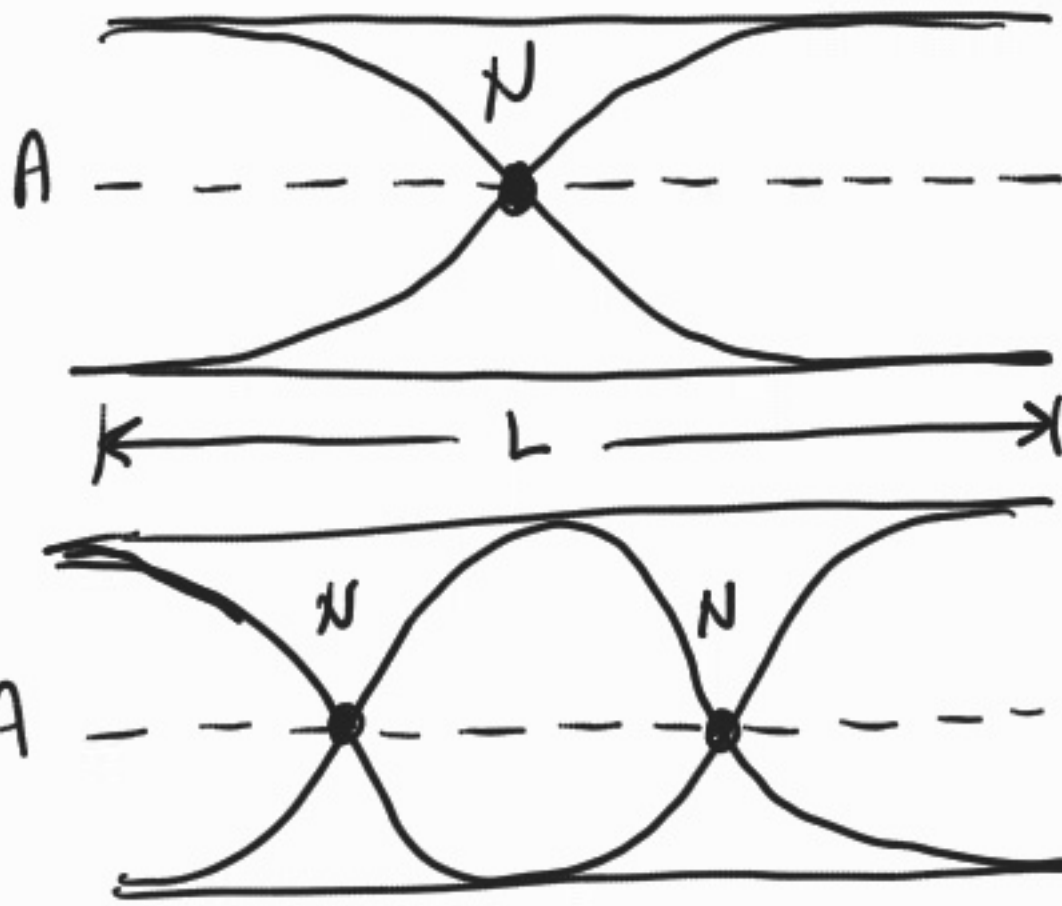


Node — Pressure ⟷ Antinode — displacement

displacement ⟷ Pressure

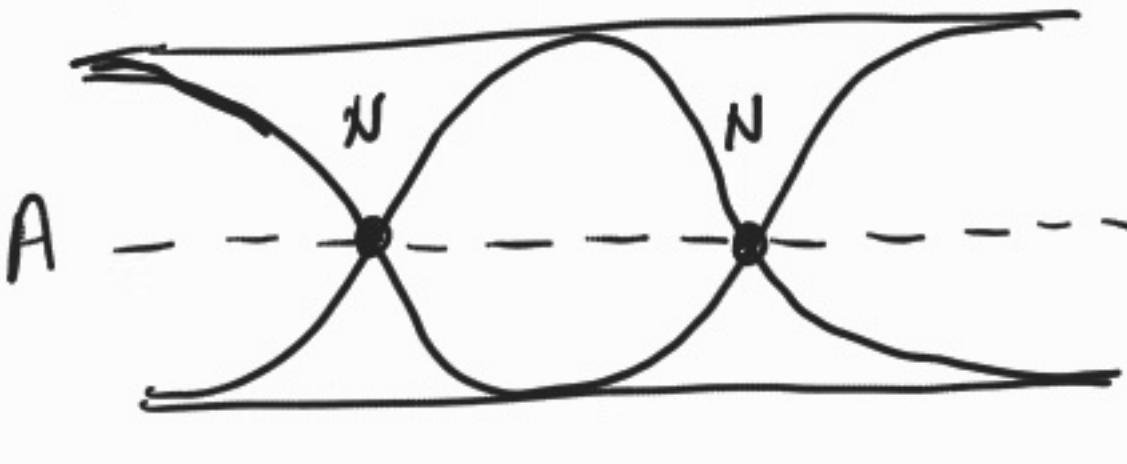
Standing Sound Waves

- Pipe with both ends open.
- Displacement profile.



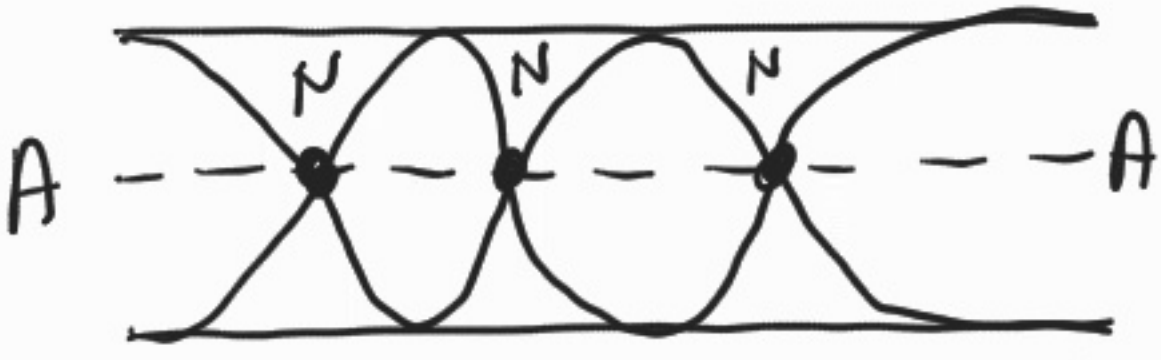
[1st Harmonic]

$\lambda = 2L$



[2nd Harmonic]

$\lambda = L$



[3rd Harmonic]

$\lambda = \frac{2}{3}L$

- Frequency of the nth harmonic:

$f_n = nV/2L$

$f_1 = V/2L$

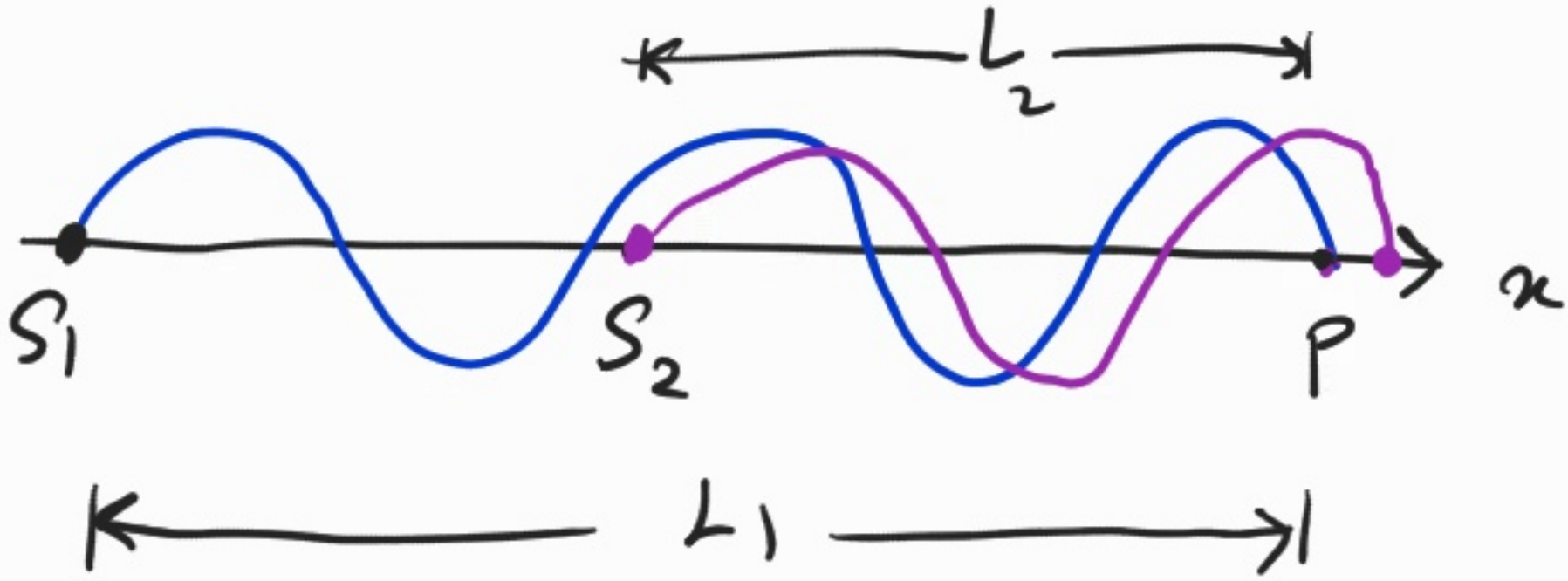
[fundamental f]

$\lambda = \frac{2L}{n}$ where

$n = 1, 2, 3, \ldots$

Interference of Sound Waves

- Consider 2 sound waves with the same amplitude travelling along the x-axis. What would the sound wave be at point P?



$$S_1 = S_m \cos(kx - \omega t + \phi_1)$$

$$S_2 = S_m \cos(kx - \omega t + \phi_2)$$

Interference of Sound Waves

- Resultant Sound wave – S_R

$$S_R = (S_1 + S_2)$$

$$= \left[2 S_m \cos\left(\frac{\Delta\phi}{2}\right)\right] \cos\left(kx - \omega t + \frac{\phi_1 + \phi_2}{2}\right)$$

amplitude of the resultant sound wave at P.

$$\Delta\phi = \phi_1 - \phi_2$$

- The amplitude is max if

$$\frac{\Delta\phi}{2} = m\pi \quad \text{or} \quad \Delta\phi = 2m\pi$$

$m = 0, 1, 2, \ldots$

[CONSTRUCTIVE INTERFERENCE]

- The amplitude is zero if:

$$\frac{\Delta\phi}{2} = \left(m + \frac{1}{2}\right)\pi \quad \text{or} \quad \Delta\phi = (2m+1)\pi$$

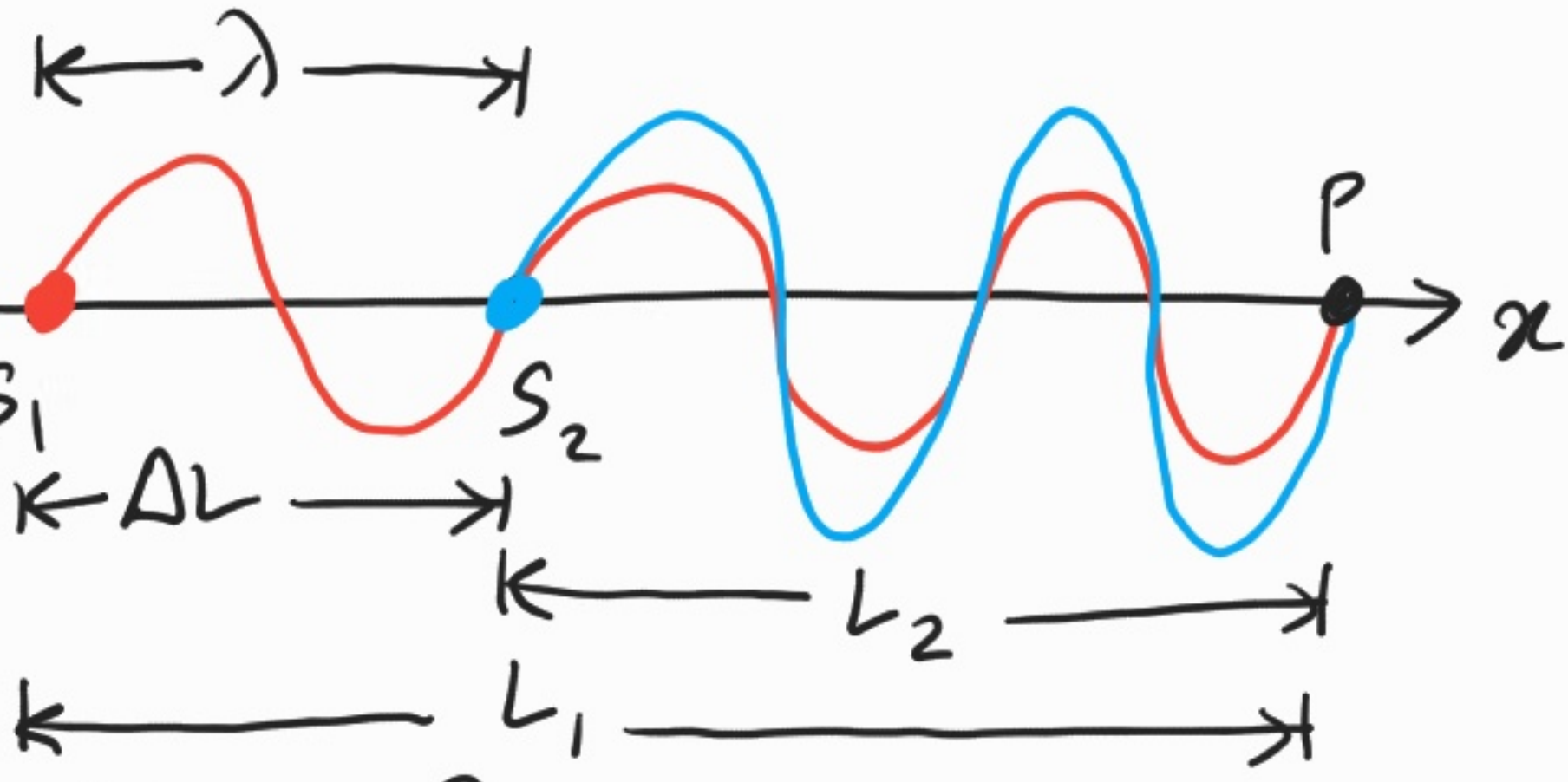
$m = 0, 1, 2, \ldots$

[DESTRUCTIVE INTERFERENCE]

Interference of Sound Waves

Alternative Analysis

[CONSTRUCTIVE INTERFERENCE]



$$\Delta L = m\lambda$$

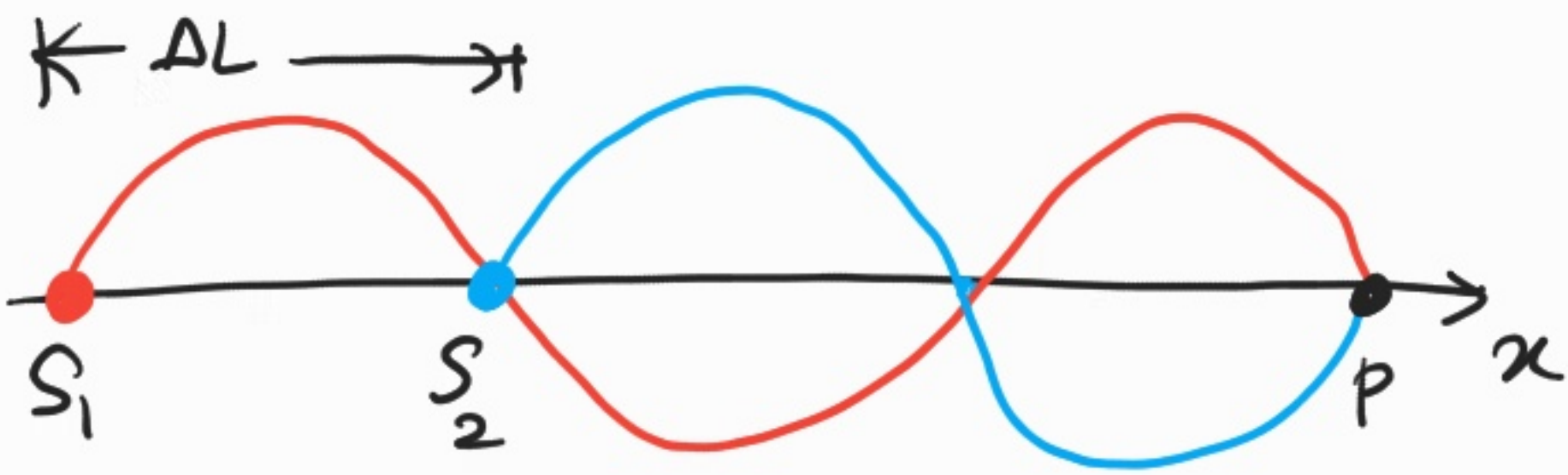
$m = 0, 1, 2, \ldots$

Interference of Sound Waves

Alternative Analysis

[DESTRUCTIVE INTERFERENCE]

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda \quad , \quad m = 0, 1, 2, \cdots$$



$$\Delta L = m\lambda$$

$$\Delta\phi = \frac{\Delta L}{\lambda} 2\pi = m 2\pi$$

$$\left[\begin{matrix}\text{Phase}\\ \text{Difference}\end{matrix}\right] = \frac{2\pi}{\lambda}\left[\begin{matrix}\text{Path Length}\\ \text{Difference}\end{matrix}\right]$$